Charge transport, not interaction range, bounds the Lieb–Schultz–Mattis twist cost

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The standard variational proof of the Lieb–Schultz–Mattis theorem twists the spins slowly around the ring and shows the twist costs almost nothing, and the standard way of pricing that twist charges each interaction for the distance it spans. The estimate then fails for the slowly decaying tails that programmable quantum simulators actually have. We show that the twist does not see geometric range alone: a term's exact cost is the spread of twist phases over the charge configurations it connects, bounded by how far it must move charge once every weighted charge it conserves is removed. This gives $\Delta_N \leq 8\pi^2 S^2 \mathcal{T}_2/L^2$ for the gap at fractional filling, with a transport moment $\mathcal{T}_2$ independent of how the Hamiltonian is written. Interactions with zero wrapped-polarization displacement cost exactly zero at any range: a spin-1/2 ring dressed with arbitrary translation-invariant diagonal tails and with correlated hoppings whose displacements cancel still obeys $\Delta_N \leq \pi^2 |J_\perp|/L$, set by its exchange alone. On a cylinder the cost resolves by direction, and a dense dipolar tail still leaves the bound vanishing when $W^2[1 + \log(L/W)] = o(L)$.

The main results of this paper were obtained through a generative-AI workflow using OpenAI GPT-5.6 Sol, Anthropic Claude Fable 5, and Grok 4.6. Further details appear in the disclosure at the end of the paper.

Introduction.—A spin chain carrying a half-odd-integer spin in each unit cell cannot have both a unique ground state and an energy gap that survives as the chain grows [1–5]. The statement is remarkable because it needs no solution of the model. One rotates the spins about the z axis by an angle that winds once around the ring, argues that this costs an energy of order $1/L$, and argues separately that the rotated state is orthogonal to the one it came from. A second state at nearly the same energy is the absence of a gap. The conclusion constrains which phases a lattice model may have at all, and it is the reason fractional filling is understood to force either gaplessness, degeneracy, or broken symmetry.

The weak point is the energy estimate. It asks what each term of the Hamiltonian feels, and answers by geometry: a term spanning r sites has its ends rotated by angles differing by $2\pi r/L$, so it is disturbed by an angle of order r/L and contributes of order $(r/L)^2$. Summing L such terms gives the familiar $O(1/L)$ when r is bounded. For a two-body tail falling off as $r^{-\alpha}$, the resulting gap bound fails to vanish unless $\alpha > 2$ [6, 7]. That is awkward, because the tails in Rydberg-dressed and polar-molecule lattices [8–10] are exactly the ones so declared dangerous, and because the estimate is visibly too pessimistic. A density–density coupling between distant sites moves no charge at all, and the twist is a phase growing with position in proportion to charge, so such a term should not notice the twist however far it reaches. The observation has a precedent: in proving the Bloch theorem for lattice models, Watanabe noted that a term crossing a twist seam is multiplied by a phase fixed by the net charge it carries across, not by its span—for short-ranged terms and a single seam [11]. Ma made the distinction quantitative for long-range couplings, separating charge-moving from charge-neutral tails and giving each its own decay threshold [6]. Zhou–Li and Liu et al. obtained complementary long-range LSM results by different methods [7, 12]. The increment here is a decomposition-independent twist-phase seminorm and a finite-angle transport proxy, together with a wrapped-polarization criterion, an exact transport moment for the XY ring, and direction-resolved cylinder and strip bounds.

We show that the twist cost is such an intrinsic quantity, and that it is bounded by intrinsic charge transport rather than geometric range alone (Fig. 1). For a single term, the exact cost is the spread of twist phases over the charge configurations that term connects. That spread is controlled by a transport distance—the displacement that remains after every weighted charge the term conserves has been subtracted—for which we give a closed formula covering correlated hopping of any order. Summing over the cheapest way of writing the Hamiltonian produces a transport moment $\mathcal{T}_2$ and the finite-ring bound $\Delta_N \leq 8\pi^2 S^2 \mathcal{T}_2/L^2$ on the gap within a charge sector. Three consequences follow. Interactions with zero wrapped-polarization displacement cost exactly zero at every range, so a spin-1/2 ring dressed with arbitrary translation-invariant diagonal tails obeys $\Delta_N \leq \pi^2 |J_\perp|/L$ fixed by its exchange coupling alone. The known $\alpha > 2$ threshold reappears as the special case in which transport happens to equal range. And on a cylinder the cost resolves by direction, so that dynamics within a transverse slice is free and a dense

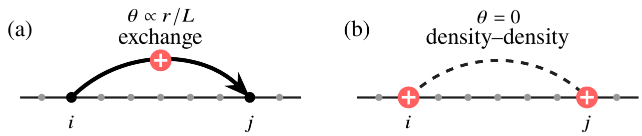

FIG. 1. Two couplings of equal span and opposite cost. (a) An exchange between sites i and j carries a charge across the intervening distance r, so the uniform twist rotates its two ends differently and it acquires a phase $\theta \propto r/L$. (b) A density–density coupling between the same two sites leaves both charges where they are; the twist phases at the two ends cancel and $\theta = 0$ exactly, at any r. The Lieb–Schultz–Mattis twist prices the first and not the second.

dipolar tail still leaves the bound vanishing whenever $W^2[1 + \log(L/W)] = o(L)$.

Setup.—Take a ring of L sites, labelled $j = 1, \ldots, L$ with site $L+1$ identified with site 1, each carrying spin S; write $\mathcal{H}_L$ for the Hilbert space of the chain. It is convenient to count charge from the bottom of the on-site multiplet, $q_j = S_j^z + S$, so that q_j takes the integer values $0, 1, \ldots, 2S$, and to write $Q = \sum_j q_j$ for the total. Nothing physical depends on the shift. Let T be translation by one site.

We assume only that the Hamiltonian conserves total charge and is invariant under translation by p sites,

$$[H_L, Q] = 0, \qquad [H_L, T^p] = 0, \tag{1}$$

so that p is the number of sites in a unit cell. Neither the individual terms of H_L nor its range enter here. Eigenvalues are ordered with multiplicity; Δ denotes the gap on the whole space, and Δ_N the gap within the sector of total charge N.

What a twist costs.—Apply the uniform twist

$$U = \exp\Big(\frac{2\pi i}{L} \sum_j j\, q_j\Big), \tag{2}$$

which rotates the spin at site j about the z axis by $2\pi j/L$, so that the rotation angle winds once around the circle as one goes once around the ring. Conjugating a term by U multiplies each of its charge-transfer processes by a phase, and the size of that phase is what the term is charged for.

Make this exact. Let B be a Hermitian term acting on a set of sites X and commuting with the charge on X. The charges $\{q_j : j \in X\}$ commute, so $\mathcal{H}_L$ splits into their joint eigenspaces, one for each assignment λ of an integer charge to every site of X; each assignment carries the twist phase

$$\theta_\lambda = \frac{2\pi}{L} \sum_{j\in X} j\lambda_j \pmod{2\pi}. \tag{3}$$

The phase θ_λ is the total angle the twist assigns to a given arrangement of charge on the sites the term touches. Now regard two assignments as linked whenever B has a nonzero matrix element between them, and for each connected family of linked assignments take the shortest arc of the circle containing its phases. Writing $\eta(B)$ for the largest radius of those arcs—the *twist-phase spread*, with $0 \le \eta(B) \le \pi$—we show in the Supplemental Material that $\eta(B)$ is the smallest angle by which any rotation built from the on-site charges must act in order to reproduce the twist on B. It is therefore a property of B alone, unchanged by any relabelling within degenerate charge eigenspaces, and it vanishes exactly when B commutes with the twist.

The spread is exact but implicit. A second quantity bounds it and can be read off a term by inspection. Collect the weighted charges that B leaves invariant,

$$\mathcal{K}_X(B) = \Big\{c \in \mathbb{R}^X : \big[B, \textstyle\sum_{j\in X} c_j S_j^z\big] = 0\Big\}, \tag{4}$$

which records every linear combination of on-site charges the term cannot change—a two-site exchange conserves the sum of its two charges but not their difference, while a density–density coupling conserves both separately. Subtracting these conserved directions from the site positions leaves the displacement the term genuinely produces. Its size is the *transport distance*

$$\tau(B) = \inf_{n\in\mathbb{Z}^X,\, c\in\mathcal{K}_X(B)} \sum_{j\in X} |j + n_j L - c_j|, \tag{5}$$

where the integers n_j allow each site to be counted around the seam of the ring, which is what makes τ independent of where the labelling starts. An exchange between two sites pays the distance between them measured around the ring; a density–density coupling of any span pays nothing, because its site positions are themselves conserved weights and the infimum is zero.

For the processes that matter in practice, τ is a formula rather than an optimization. Suppose every charge-transfer process in B moves charge in the same pattern, so the transfers are integer multiples of a single primitive vector d, and let x list the positions of the sites in X. Then

$$\tau(B) = \frac{\mathrm{dist}(d\cdot x,\, L\mathbb{Z})}{\|d\|_\infty}. \tag{6}$$

A term therefore costs what its charge displacement $d\cdot x$ falls short of a whole number of turns around the ring. For spin-1/2 the entries of d are 0 or ±1, so such a term is free exactly when the displacement $\sum_j d_j j$ is a multiple of L. This classifies correlated hopping at any order, including processes that wrap the seam.

The finite-ring bound.—A Hamiltonian has many decompositions into terms, and they do not cost the same, so the cost of H_L is the cheapest of them. Define the transport moment

$$\mathcal{T}_2(H_L) = \inf_{H_L=\sum_a B_a} \sum_a \tau(B_a)^2 \|B_a\|, \tag{7}$$

the infimum running over finite decompositions into Hermitian terms that each conserve the charge on their own support. Taking the infimum is what makes $\mathcal{T}_2$ a property of the operator rather than of a presentation of it, and it is not a formality: for the spin-1/2 XY ring the bond decomposition attains $\mathcal{T}_2 = L|J|/2$, whereas writing the same Hamiltonian as a single ring-wide term costs at least $|J|\lfloor L^2/4\rfloor^2$. Replacing τ by the sharper phase spread gives a companion quantity

$$\mathcal{B}(H_L) = \inf_{H_L=\sum_a B_a} \sum_a \sin^2\!\big(\min\{\eta(B_a), \tfrac{\pi}{2}\}\big)\|B_a\|, \quad (8)$$

which charges each term for the sine of its phase spread instead of for the spread itself, and so keeps the finite-angle cancellations that a quadratic estimate throws away.

The main result is that these quantities bound the gap. Whenever the filling is fractional, meaning that pN/L is not an integer,

$$\Delta_N(H_L) \;\le\; 2\mathcal{B}(H_L) \;\le\; \frac{8\pi^2 S^2}{L^2}\,\mathcal{T}_2(H_L). \quad (9)$$

The gap within a sector of fractional filling is therefore no larger than the Hamiltonian's transport moment divided by L^2; any family with $\mathcal{T}_2 = o(L^2)$ has vanishing gaps in every sequence of fractional sectors. The statement needs neither a unique ground state nor prior knowledge of which sector the ground state occupies, which makes it directly usable in canonical simulations at fixed particle number or magnetization. The full gap follows as a corollary: if the global ground energy is degenerate then $\Delta = 0$ outright, and if the ground state is unique and fractional then $\Delta \le \Delta_N$. For a globally SU(2)-invariant H_L with pS not an integer, uniqueness would force a singlet, whose shifted charge produces the required twist phase, so the full-gap statement holds there too.

Interactions that cost nothing.—The bound is only as interesting as the class of terms it exempts, and that class is large. Consider a spin-1/2 ring with $L \ge 6$, one site per cell, and real $J_\perp$,

$$H_L = J_\perp \sum_j \big(S^x_j S^x_{j+1} + S^y_j S^y_{j+1}\big) + V + Z, \quad (10)$$

where V is any Hermitian translation-invariant operator diagonal in the product S^z basis and Z is defined below. Being diagonal, V conserves every on-site charge separately, so $\tau(V) = 0$ term by term and V drops out of Eq. (9) entirely. It may contain many-body terms and arbitrary two-body tails,

$$V = \sum_j \sum_{r=1}^{\lfloor (L-1)/2\rfloor} V_r S^z_j S^z_{j+r} - h\sum_j S^z_j, \quad (11)$$

with no decay condition whatever on V_r: dipolar, van der Waals, or unscreened Coulomb tails all cost the same, namely nothing.

The second free class is off diagonal, which is the more surprising half. Let Z be any Hermitian translation-invariant sum of fixed-transfer spin monomials, kept as separate terms, whose charge displacement $d\cdot x$ is a multiple of L; by Eq. (6) every such term is free regardless of its order, amplitude, or span. A four-body example is, for $2 \le r \le \lfloor L/2\rfloor$,

$$C_{j,r} = g_r S^+_j S^-_{j+1} S^-_{j+r} S^+_{j+r+1} + \text{H.c.}, \quad (12)$$

in which one charge hops right across one bond while a second hops left across a distant bond. The transfer vector $(1,-1,-1,1)$ on positions $(j, j{+}1, j{+}r, j{+}r{+}1)$ has vanishing first moment, so the two hops move charge in opposite senses by equal amounts, the twist phases cancel exactly, and $\tau(C_{j,r}) = 0$ even when the pair straddles the seam. The two charges move; the polarization does not.

What remains is the exchange. Every free transfer above annihilates the one-particle sector, while each transverse bond has norm $|J_\perp|/2$ and transport distance 1; matching one-particle matrix elements against every admissible decomposition (see the Supplemental Material) gives $\mathcal{T}_2 = L|J_\perp|/2$ exactly. The physical bond decomposition also gives $\mathcal{B}(H_L) \le (L|J_\perp|/2)\sin^2(\pi/L)$ for $L \ge 6$, so

$$\Delta_N(H_L) \le L|J_\perp|\sin^2\frac{\pi}{L} \le \frac{\pi^2|J_\perp|}{L}, \qquad N/L \notin \mathbb{Z}, \quad (13)$$

the first form being strictly sharper at any nonzero $J_\perp$. The bound is set by the exchange alone, independent of the strength, sign, range, and decay of everything in V and Z. What is exact here is the quadratic transport moment; the finite-angle expression is an explicit upper bound on the twist cost, and Eq. (13) is an upper bound on the gap rather than an evaluation of it.

The mechanism survives a genuine unit cell and chiral hopping. With the transverse term replaced by p-periodic couplings $\sum_j [J_j(S^x_j S^x_{j+1} + S^y_j S^y_{j+1}) + D_j(S^x_j S^y_{j+1} - S^y_j S^x_{j+1})]$, p dividing L and the free terms also p-periodic, every fractional sector obeys $\Delta_N \le (L/p)\sin^2(\pi/L)\sum_a \sqrt{J_a^2 + D_a^2} \le (\pi^2/pL)\sum_a \sqrt{J_a^2 + D_a^2}$, covering uniform and dimerized XXZ chains with Dzyaloshinskii–Moriya coupling [13]. At $p = 2$ and half filling pN/L is an integer and the hypothesis correctly fails, consistent with the symmetry-allowed dimerization gap.

Cylinders and ladders.—Because the twist is applied along one direction only, the cost it assigns is directional, and this survives on a cylinder (Fig. 2): L slices around a periodic direction, each slice a copy of a graph with W sites carrying integer charges, with the twist applied to the periodic coordinate alone. Choose one oriented representative e of each hopping bond, running from a site in slice m to a site in slice $m + r_e$, with amplitude t_e, and write ℓ_e for the distance from r_e to the nearest multiple of

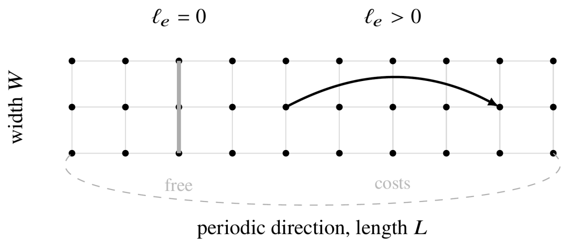

FIG. 2. Unrolled cylinder: L slices around the periodic direction, each of width W. The longitudinal twist charges a hopping bond by its wrapped displacement ℓ_e, so a bond confined to one transverse slice, with $\ell_e = 0$, costs nothing regardless of its strength.

L. If the bonds and amplitudes are p-periodic and the remaining charge-conserving dynamics commutes with its own longitudinal twist, then every fractional sector obeys

$$\Delta_N(H_L) \le 2\sum_e |t_e| \sin^2 \frac{\pi \ell_e}{L}. \tag{14}$$

Only the longitudinal displacement appears: a bond inside one transverse slice has $\ell_e = 0$ and costs exactly zero, however strong and however many sites of the slice it couples. For uniform nearest-neighbor hopping this is at most $2\pi^2 W|t|/L$, vanishing for bounded hopping whenever $W = o(L)$.

Long-range hopping enters only through how much amplitude moves charge a given longitudinal distance, so a profile $A_L(r) \le CLK_L r^{-\alpha}$ with $\alpha > 2$ for the total amplitude $A_L(r)$ of the bonds with $\ell_e = r$ leaves the bound vanishing whenever $K_L = O(L^\beta)$ with $\beta < \min\{1, \alpha - 2\}$. Density of channels is not what matters. The sharpest case is geometrically dense: on an $L \times W$ strip with every longitudinal amplitude bounded by $J(r_e^2 + s_e^2)^{-\alpha/2}$, s_e being the transverse distance, summing the Euclidean tail before applying Eq. (14) gives, for $W \le \lfloor L/2 \rfloor$, at dipolar decay $\alpha = 3$

$$\Delta_N(H_L) \le \frac{2\pi^2 J W^2}{L}\left[3 + \log\frac{\lfloor L/2\rfloor}{W}\right]. \tag{15}$$

Even with every pair of transverse channels coupled by a dipolar exchange, the sector gap is still forced to vanish provided $W^2[1 + \log(L/W)] = o(L)$; a van der Waals tail gives $O(W/L)$. This is a statement about long thin strips, not an isotropic two-dimensional theorem. At fixed width it yields a ladder corollary: for even L and an odd number of legs, a spin-1/2 SU(2)-invariant ladder has $\Delta \le \pi^2 \sum_a |J_a|/L$ whatever SU(2)-invariant dynamics acts within a rung, that dynamics being longitudinally free. Higher-dimensional LSM bounds on tori are known [14], and fixed-width ladder results are known [5, 15]; new here is the direction-resolved cost and the crossover it gives as the width grows.

Why it works.—Termwise symmetry is manufactured rather than assumed: averaging any decomposition $H_L = \sum_X \Phi_X$ over the local charge rotations, $B_X = \frac{1}{2\pi}\int_0^{2\pi} e^{i\vartheta Q_X}\Phi_X e^{-i\vartheta Q_X} d\vartheta$, preserves the sum, the supports, and Hermiticity, does not increase norms, and leaves each term commuting with the charge on its own support. Global charge conservation is the only input. The twist acting on such a term is then replaced by the smallest rotation built from on-site charges that reproduces it, which is what $\eta(B_X)$ measures; the transport distance supplies an explicit one, generated by $\frac{2\pi}{L}\sum_{j\in X}(j + n_j L - c_j) S_j^z$ with norm $2\pi S\tau(B_X)/L$. This is the step that converts geometry into transport, and it gives $\eta(B) \le \min\{2\pi S\tau(B)/L, \pi\}$.

Twisting in both senses and averaging then cancels the linear response, so that for self-adjoint A and B

$$\begin{aligned}\left\|\tfrac{1}{2}\left(e^{-iA}Be^{iA} + e^{iA}Be^{-iA}\right) - B\right\| \\ \le 2\sin^2\!\left(\min\{\|A\|, \tfrac{\pi}{2}\}\right)\|B\| :\end{aligned} \tag{16}$$

the cost is second order in the angle—the familiar reason a twist is cheap—but held at finite angle rather than expanded, which is what keeps Eq. (9) sharper than its quadratic form. Orthogonality closes the argument. The twist commutes with total charge, so the twisted states stay in the sector, while $T^p U T^{-p} = U e^{-2\pi i p Q/L}$ shows that at fractional filling a sector ground state and its two twists carry different translation eigenvalues and cannot coincide. One of the twists therefore lies within the paired average of the ground energy, and the variational principle within the sector gives Eq. (9).

Discussion.—The functional reproduces what is known wherever transport and range happen to agree. Bounding τ by the spread of a term's support recovers the finite-range theorem with an explicit constant—if every support has diameter at most R, $L > 3R$, and the local strength at most J, then $\Delta_N \le 8\pi^2 S^2 \kappa_R^2 J/L$ with $\kappa_R = \lfloor (R+1)^2/4 \rfloor$—and for power-law couplings it returns the threshold $\alpha > 2$ when charge moves and the far weaker $\alpha > 0$ when it does not [6]. Equation (9) is never weaker than the geometric estimate and is strictly sharper whenever weighted charges beyond the total are conserved. The difference is not subtle: for a V/r^α density tail, support-based accounting adds $\frac{\pi^2|V|}{2L}\sum_r r^{2-\alpha}$, of order $\log L/L$ at dipolar decay, where the true contribution is zero.

Two limits deserve emphasis. These are upper bounds on the gap, not evaluations of it: a vanishing bound forbids a uniform gap but says nothing about the true low-energy scale, and the diagonal tails that drop out of the estimate may well dominate the energy and select the phase. And the cylinder results are anisotropic, requiring the width to grow more slowly than the circumference; they do not settle the isotropic two-dimensional problem.

The practical reading is that the interactions available on programmable platforms are far less dangerous to the obstruction than range-based accounting suggests. Rydberg-dressed extended Bose–Hubbard chains have been realized [10], and dipolar spin exchange observed in molecular lattices and tweezer arrays [16–18]; for a spin-1/2 chain, diagonal tails of any decay drop out of the estimate altogether, as do the correlated hoppings dipole-conserving models generate [19] when their displacements cancel. We do not claim any experiment realizes the hypotheses used here; the transport functional is the tool for checking. The relevant question for a long-range coupling is how its charge-transfer processes couple to the twist, rather than its geometric span alone. Related ideas appear in the many-body charge-transport index [20]; the object here acts directly on a finite-volume Hamiltonian and needs no global dipole conservation.

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Disclosure of automated assistance.—OpenAI GPT-5.6 Sol, Anthropic Claude Fable 5, and Grok 4.6 were used extensively throughout the development and preparation of this manuscript. The generative-AI workflow developed and refined the physical interpretation of charge transport in the LSM twist construction, checked analytical derivations and limiting regimes, constructed and tested the spin-chain, ladder, cylinder, and long-range-interaction examples, compared the results with prior literature, designed and revised the schematic figures, and organized, drafted, and revised the REVTeX manuscript. The results await further verification by domain experts.

Data availability.—No data were created or analyzed. All results are analytic.

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Supplemental Material for "Charge transport, not interaction range, bounds the Lieb–Schultz–Mattis twist cost"

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In this Supplemental Material, we provide detailed derivations of the following topics: the reduction of the twist cost of a single interaction term to an exact charge-algebra rotation, the sharp finite-angle inequality and the orthogonality supplied by the translation phase, the seminorm and duality properties of the Hamiltonian transport costs together with the exact evaluation of the XY ring, the transfer-matrix certificates and the closed formula for correlated hopping, the recovery of the finite-range theorem and of the known power-law decay thresholds, the direction-resolved bounds on cylinders and ladders, and the spin-1/2 ring whose diagonal tails and polarization-neutral hoppings cost exactly nothing.

I. STATEMENT OF RESULTS

We follow the notation and hypotheses of the main text throughout, and work at fixed ring length $L \geq 2$. Operator norms are norms on the full Hilbert space; a supported decomposition is finite, with each term Hermitian and acting trivially outside its support, and repeated support labels are permitted.

Besides the twist-phase spread $\eta(B)$ and the transport distance $\tau(B)$, the proofs below use the finite-angle cost each one assigns to a single term,

$$s(B) = \sin\Big(\min\Big\{\eta(B), \frac{\pi}{2}\Big\}\Big), \qquad s_\tau(B) = \sin\Big(\min\Big\{\frac{2\pi S}{L}\tau(B), \frac{\pi}{2}\Big\}\Big), \tag{S1}$$

so that $\mathcal{B}(H_L) = \inf \sum_a s(B_a)^2 \|B_a\|$ is the quantity of the main text, and the transport proxy defines its companion

$$\mathcal{B}_\tau(H_L) = \inf_{H_L = \sum_a B_a} \sum_a s_\tau(B_a)^2 \|B_a\|, \tag{S2}$$

the infimum running, as for $\mathcal{B}$ and $\mathcal{T}_2$, over finite supported Hermitian decompositions whose terms conserve the charge on their own supports.

The results. We prove the following four statements.

Result A (bound for a given decomposition). Let H_L satisfy the symmetry assumptions of the main text, take any decomposition $H_L = \sum_X \Phi_X$ into Hermitian supported terms — with no symmetry assumed of the individual Φ_X — and let B_X be its charge average, Eq. (S5). Then for every charge N with pN/L not an integer,

$$\Delta_N(H_L) \leq 2\sum_X s(B_X)^2 \|B_X\| \leq 2\sum_X s_\tau(B_X)^2 \|B_X\| \leq \frac{8\pi^2 S^2}{L^2} \sum_X \tau(B_X)^2 \|B_X\|. \tag{S3}$$

Each inequality trades exactness for computability: the first cost is the true one, the second replaces it by transported charge, the third expands the sine.

Result B (bound for the Hamiltonian). Under the same hypotheses,

$$\Delta_N(H_L) \leq 2\mathcal{B}(H_L) \leq 2\mathcal{B}_\tau(H_L) \leq \frac{8\pi^2 S^2}{L^2} \mathcal{T}_2(H_L). \tag{S4}$$

This is the central bound of the main text, and it depends on no choice of presentation.

Result C (the full gap). If the global ground energy is degenerate then $\Delta(H_L) = 0$. If the global ground state is unique and carries a charge N with pN/L not an integer, then $\Delta(H_L) \leq \Delta_N(H_L)$ and Eq. (S4) bounds the full gap.

Result D (half-odd-integer spin). If H_L commutes with the global SU(2) action and pS is not an integer, the full-gap bound of Result C holds. Half-odd-integer spin with one-site translation is the special case $p = 1$.

II. TERMWISE CHARGE CONSERVATION FROM THE GLOBAL SYMMETRY

The main text assumes only that the *Hamiltonian* conserves charge. Every later step, however, works with terms that conserve the charge on their own supports. This section closes that gap: any decomposition can be replaced by one with termwise conservation, keeping the sum, the supports, and the Hermiticity, and without increasing any norm. Nothing is assumed about the interaction beyond the global symmetry.

Average each term over the rotations generated by the charge on its own support,

$$B_X = \frac{1}{2\pi}\int_0^{2\pi} e^{i\vartheta Q_X}\Phi_X e^{-i\vartheta Q_X}\,d\vartheta, \qquad Q_X = \sum_{j\in X} q_j. \tag{S5}$$

This is the projection onto the part of Φ_X that commutes with Q_X; the integral runs over a compact group, so it is a unital completely positive map and is contractive. The contraction also follows without that structure: every conjugate $e^{i\vartheta Q_X}\Phi_X e^{-i\vartheta Q_X}$ has the same norm as Φ_X, so the triangle inequality for the integral gives $\|B_X\| \le \|\Phi_X\|$.

The averages still sum to H_L. Because Φ_X acts trivially outside X, the charge outside X commutes with it and

$$e^{i\vartheta Q_X}\Phi_X e^{-i\vartheta Q_X} = e^{i\vartheta Q}\Phi_X e^{-i\vartheta Q}, \tag{S6}$$

so the local average is the global one in disguise. Summing over X and using $[H_L, Q] = 0$,

$$\sum_X B_X = \frac{1}{2\pi}\int_0^{2\pi} e^{i\vartheta Q} H_L e^{-i\vartheta Q}\,d\vartheta = H_L. \tag{S7}$$

Hermiticity and support are manifestly preserved, and expanding Φ_X into eigenoperators of ad_{Q_X} shows that the average keeps only the piece of charge transfer zero, so $[B_X, Q_X] = 0$. Termwise conservation is therefore a consequence of the global hypothesis, not an extra condition on the interaction.

III. LOCAL REDUCTION OF THE GLOBAL TWIST

The twist

$$U = \exp\Big(\frac{2\pi i}{L}\sum_j j\, q_j\Big) \tag{S8}$$

is a global object, but a term feels it only through the sites it touches. This section makes that restriction explicit and shows that the effective local generator can be shifted by any weighted charge the term conserves — which is the freedom the transport distance later exploits.

Let B be supported on X with $[B, Q_X] = 0$. Restricted to X, conjugation by U is generated by

$$G_X = \frac{2\pi}{L}\sum_{j\in X} j\, q_j. \tag{S9}$$

Because each q_j has integer spectrum, adding L to any coefficient changes e^{iG_X} by $e^{2\pi i q_j} = \mathbb{1}$; hence for every integer vector $n \in \mathbb{Z}^X$,

$$e^{iG_X} = \exp\Big[\frac{2\pi i}{L}\sum_{j\in X}(j + n_j L)\, q_j\Big]. \tag{S10}$$

Each site may therefore be counted around the seam of the ring at will, which is why nothing below depends on where the labelling starts.

The second freedom is the conserved weights. Let

$$\mathcal{K}_X(B) = \Big\{c \in \mathbb{R}^X : \big[B, \textstyle\sum_{j\in X} c_j S_j^z\big] = 0\Big\} \tag{S11}$$

be the weighted charges the term leaves invariant. For $c \in \mathcal{K}_X(B)$ the operator $C = \sum_j c_j S_j^z$ commutes with B and with every q_j, so it may be subtracted from the generator without changing the conjugation:

$$e^{-iG_X} B e^{iG_X} = e^{-iA} B e^{iA}, \qquad A = A_X(n, c) = \frac{2\pi}{L}\sum_{j\in X}(j + n_j L - c_j) S_j^z. \tag{S12}$$

The replacement of q_j by S^z_j shifts the generator by a multiple of the identity, which drops out of any conjugation. Equation (S12) is the technical heart of the paper: the twist acting on B is a rotation whose angle is not the span of X but whatever remains after the conserved weights are removed.

The norm of that generator is exact rather than estimated. For real coefficients a_j the commuting operator $\sum_j a_j S^z_j$ has eigenvalues $\sum_j a_j m_j$ with each $m_j \in \{-S, \dots, S\}$ selectable independently, and choosing the extremal sign at every site gives

$$\Big\|\sum_j a_j S^z_j\Big\| = S\sum_j |a_j|, \qquad \text{hence} \qquad \|A_X(n,c)\| = \frac{2\pi S}{L}\sum_{j\in X}|j + n_j L - c_j|. \tag{S13}$$

No minimizing pair (n, c) need exist: every estimate below holds for each admissible pair and therefore passes to the infimum defining $\tau(B)$.

IV. EXACT TERM COST FROM THE JOINT-CHARGE PHASE GRAPH

Section III produced *some* local rotation reproducing the twist. The cost of a term is set by the *smallest* one, and this section identifies it exactly. The answer is a property of the term alone: the spread of twist phases across the charge configurations the term connects. The construction uses only that the on-site charges are commuting operators with integer spectrum, so it covers arbitrary finite-dimensional sites and degenerate charge eigenspaces.

For each assignment $\lambda = (\lambda_j)_{j\in X}$ of an integer charge to every site of X, let $\mathcal{P}_\lambda$ be the joint spectral projector and θ_λ the twist phase it carries,

$$\mathcal{P}_\lambda = \prod_{j\in X} \mathbb{1}_{\{\lambda_j\}}(q_j), \qquad \theta_\lambda = \frac{2\pi}{L}\sum_{j\in X} j\lambda_j \pmod{2\pi}, \tag{S14}$$

so that the local twist is the diagonal unitary $V_X = \sum_\lambda e^{i\theta_\lambda}\mathcal{P}_\lambda$. Join λ to μ whenever $\mathcal{P}_\lambda B \mathcal{P}_\mu \neq 0$, that is, whenever the term actually connects the two configurations. For each connected component ω of the resulting graph let r_ω be the radius of the shortest arc of the circle containing its phases,

$$r_\omega = \inf_{\alpha\in\mathbb{R}} \max_{\lambda\in\omega} d_{\mathbb{T}}(\theta_\lambda, \alpha), \qquad \eta(B) = \max_\omega r_\omega \in [0, \pi], \tag{S15}$$

with $d_{\mathbb{T}}$ the distance along the circle. Using spectral projectors rather than individual basis vectors makes $\eta(B)$ invariant under any relabelling inside a degenerate charge eigenspace.

The spread as the minimal rotation. Let $\mathcal{A}_X$ be the abelian algebra generated by the charges $\{q_j\}_{j\in X}$, whose self-adjoint elements are exactly $A = \sum_\lambda a_\lambda \mathcal{P}_\lambda$ with real a_λ. Then

$$\eta(B) = \min\big\{\|A\| : A = A^\dagger \in \mathcal{A}_X,\ e^{-iA}Be^{iA} = V_X^{-1}BV_X\big\}. \tag{S16}$$

The spread $\eta(B)$ is thus the smallest angle through which a rotation built from the on-site charges must act in order to reproduce the twist on B.

Proof. For the upper bound, pick for each component ω a centre α_ω of a shortest containing arc, and real lifts β_λ of θ_λ with $|\beta_\lambda - \alpha_\omega| \le r_\omega$. Set $A = \sum_\lambda(\beta_\lambda - \alpha_{\omega(\lambda)})\mathcal{P}_\lambda$, so $\|A\| \le \eta(B)$. The unitary $D = \sum_\lambda e^{-i\alpha_{\omega(\lambda)}}\mathcal{P}_\lambda$ is constant on every component, hence commutes with B, and $e^{iA} = DV_X$; the two conjugations therefore agree.

Conversely, suppose $A \in \mathcal{A}_X$ is self-adjoint with $e^{-iA}Be^{iA} = V_X^{-1}BV_X$. Then $W = e^{iA}V_X^{-1}$ commutes with B. Since W is a scalar on each joint charge subspace, every nonzero block $\mathcal{P}_\lambda B\mathcal{P}_\mu$ forces those two scalars to agree, so W is constant on each component. The coefficients a_λ are therefore real lifts of that component's twist phases up to one common circular shift, whence $\max_{\lambda\in\omega}|a_\lambda| \ge r_\omega$ and $\|A\| \ge \eta(B)$. □

Combining Eq. (S16) with the paired-twist inequality of Sec. V gives the exact term estimate

$$\Big\|\tfrac{1}{2}\big(V_X^{-1}BV_X + V_XBV_X^{-1}\big) - B\Big\| \le 2s(B)^2\|B\|, \tag{S17}$$

the reverse conjugation being covered because A, V_X and W all lie in the same abelian algebra and W commutes with B.

Transport as an upper bound on the spread. Every admissible pair (n, c) in Sec. III supplies an element of $\mathcal{A}_X$ reproducing the twist, so Eqs. (S13) and (S16) give

$$\eta(B) \le \min\Big\{\frac{2\pi S}{L}\tau(B), \pi\Big\}, \qquad \text{hence} \qquad s(B) \le s_\tau(B). \tag{S18}$$

For general integer charges the same construction, after centring each q_j, gives

$$\eta(B) \le \frac{\pi}{L}\inf_{n,c}\sum_{j\in X} w_j|j + n_jL - c_j|, \qquad w_j = \max\operatorname{spec}(q_j) - \min\operatorname{spec}(q_j), \tag{S19}$$

where $\sum_j c_j q_j$ commutes with B; the weight w_j is the range of charge available at site j.

Criterion for zero cost. The graph gives the exact criterion

$$\eta(B) = 0 \iff [B, V_X] = 0, \tag{S20}$$

since either statement says the twist phase is constant on every connected component. For a spin-1/2 term whose processes all carry the same transfer vector d, the components are pairs and

$$\eta(B) = \frac{\pi}{L}\operatorname{dist}\Big(\sum_j d_j j,\ L\mathbb{Z}\Big), \tag{S21}$$

which agrees with the transport formula of Sec. VIII. At higher spin the two can differ, and the phase spread is the sharper of the two. Take $L = 6$, two spin-1 sites at positions 0 and 3, and

$$B = |{+1}, {-1}\rangle\langle {-1}, {+1}| + |{-1}, {+1}\rangle\langle {+1}, {-1}|. \tag{S22}$$

The actual transfer is $(2, -2)$, accumulating a phase 2π, so $\eta(B) = 0$ and the term is free. The transport distance, built from the real span of the transfer rows, sees only the primitive direction $(1, -1)$ and returns $\tau(B) = 3$. The phase spread therefore detects an exact cancellation that its computable proxy misses.

V. THE PAIRED-TWIST INEQUALITY

The remaining ingredient is quantitative: how much energy a rotation of a given angle can extract from a term. Twisting in both senses and averaging removes the linear response, leaving a cost second order in the angle — the familiar reason a slow twist is cheap — and the following inequality holds that statement at finite angle instead of expanding it.

For self-adjoint A and B,

$$\Big\|\tfrac{1}{2}\big(e^{-iA}Be^{iA} + e^{iA}Be^{-iA}\big) - B\Big\| \le 2\sin^2\Big(\min\Big\{\|A\|, \frac{\pi}{2}\Big\}\Big)\|B\|. \tag{S23}$$

Proof. Write $\mathcal{C}(B) = e^{-iA}Be^{iA}$ for the conjugation, which is an isometry. The paired average factorizes as

$$\frac{\mathcal{C} + \mathcal{C}^{-1}}{2} - I = \frac{1}{2}\mathcal{C}^{-1}(\mathcal{C} - I)^2, \tag{S24}$$

which is the algebraic statement that the linear term cancels. Put $a = \|A\|$ and $V = e^{-iA}$. For $a \le \pi/2$, functional calculus and $|e^{-it} - \cos a| \le \sin a$ for $|t| \le a$ give $\|V - \cos(a)I\| \le \sin a$, hence

$$\|(\mathcal{C} - I)(B)\| = \|[V, B]\| \le 2\sin(a)\|B\|, \tag{S25}$$

where subtracting the scalar $\cos(a)I$ is legitimate because it commutes with B. For $a \ge \pi/2$ the trivial bound $\|\mathcal{C} - I\| \le 2$ applies. Substituting into Eq. (S24) gives Eq. (S23). □

The coefficient cannot be improved. With $A_a = a\operatorname{diag}(1, -1)$ and $B = \sigma_x$, direct computation gives

$$\Big\|\tfrac{1}{2}\big(e^{-iA_a}Be^{iA_a} + e^{iA_a}Be^{-iA_a}\big) - B\Big\| = 1 - \cos(2a) = 2\sin^2 a, \qquad 0 \le a \le \frac{\pi}{2}, \tag{S26}$$

and such a two-level block can be embedded between product states of equal total S^z, so termwise charge conservation alone does not improve it.

Chaining Eq. (S23) through the exact logarithm and then through transport gives the term-by-term estimate that Result A sums,

$$\Big\|\tfrac{1}{2}\big(U^{-1}B_XU + UB_XU^{-1}\big) - B_X\Big\| \le 2s(B_X)^2\|B_X\| \le 2s_\tau(B_X)^2\|B_X\| \le \frac{8\pi^2S^2}{L^2}\tau(B_X)^2\|B_X\|, \tag{S27}$$

the last step being $\sin^2(\min\{x, \pi/2\}) \le x^2$ with $x = 2\pi S\tau(B_X)/L$.

VI. TRANSLATION PHASE, ORTHOGONALITY, AND THE VARIATIONAL BOUND

A cheap state proves nothing unless it is orthogonal to the ground state. Orthogonality here is exact and comes from a phase the twist picks up under translation, not from any property of the ground state itself.

Reindexing Eq. (S8) using $Tq_jT^{-1} = q_{j+1}$ and $e^{2\pi i q_j} = \mathbb{1}$ gives

$$TUT^{-1} = Ue^{-2\pi iQ/L}, \qquad \text{and iterating,} \qquad T^pUT^{-p} = Ue^{-2\pi ipQ/L}. \tag{S28}$$

Within a sector of charge N the twist thus multiplies the translation eigenvalue by $e^{-2\pi ipN/L}$, which is not 1 precisely when pN/L is not an integer — the fractional-filling condition, arriving here as an exact algebraic statement rather than an estimate.

Fix such a sector. Both H_L and T^p preserve it, and U commutes with Q, so $U^{\pm 1}\Psi$ stays in the sector for any Ψ in it. If the sector ground state Ψ is unique it is a T^p eigenvector, and by Eq. (S28) the states $U^{\pm 1}\Psi$ carry a different T^p eigenvalue; they are therefore exactly orthogonal to Ψ. Writing

$$e_\sigma = \langle U^\sigma \Psi, H_L U^\sigma \Psi\rangle - E_{0,N}, \qquad \sigma = \pm 1, \tag{S29}$$

both are nonnegative, and taking the ground-state expectation of Eq. (S27) summed over X gives

$$\frac{e_+ + e_-}{2} \le 2\sum_X s(B_X)^2 \|B_X\|. \tag{S30}$$

At least one sign lies below the right-hand side, and that twisted state is a normalized trial state orthogonal to Ψ inside the sector; the variational principle restricted to the sector then gives Result A. If the sector ground energy is degenerate the conclusion is automatic, since $\Delta_N = 0$. A fractional sector always has dimension at least two, because the only one-dimensional charge sectors put every site in its lowest or highest charge state, for which pN/L is an integer.

For Result D, global SU(2) invariance and uniqueness force the ground state to be a singlet, so $N = LS$; its translation phase is nontrivial exactly when pS is not an integer. If LS is not an integer a singlet is impossible outright, since Q has integer spectrum, and Result C applies.

VII. HAMILTONIAN TRANSPORT SEMINORMS AND DUALITY CERTIFICATES

Result A holds for every admissible decomposition, and the one-term decomposition H_L itself is always admissible. Taking infima therefore gives Result B immediately. What this section establishes is that the resulting quantities behave like norms, and that each is the best possible quantity with its one-term bound — so nothing sharper can be extracted from term-by-term estimates alone.

Seminorm properties. If $H = \sum_a B_a$ and $K = \sum_b C_b$ are admissible, their union is an admissible decomposition of $H + K$, so all three quantities are subadditive:

$$\mathcal{B}(H+K) \le \mathcal{B}(H) + \mathcal{B}(K), \qquad \mathcal{B}_\tau(H+K) \le \mathcal{B}_\tau(H) + \mathcal{B}_\tau(K), \qquad \mathcal{T}_2(H+K) \le \mathcal{T}_2(H) + \mathcal{T}_2(K). \tag{S31}$$

For a nonzero real μ the conserved weights and the phase graph are unchanged, $\mathcal{K}_X(\mu B) = \mathcal{K}_X(B)$ and $\Gamma_{\mu B} = \Gamma_B$, while $\|\mu B\| = |\mu|\|B\|$; scaling decompositions both ways gives exact homogeneity,

$$\mathcal{B}(\mu H) = |\mu|\mathcal{B}(H), \qquad \mathcal{B}_\tau(\mu H) = |\mu|\mathcal{B}_\tau(H), \qquad \mathcal{T}_2(\mu H) = |\mu|\mathcal{T}_2(H). \tag{S32}$$

They are seminorms rather than norms precisely because they vanish on nonzero operators that commute with the twist, which is the phenomenon reported in the main text. They are independent of any chosen presentation of the interaction, but remain finite-volume quantities tied to the fixed lattice and on-site charge.

Maximality among seminorms with the same one-term bound. Let p_* be any seminorm on globally U(1)-invariant Hermitian operators satisfying the one-term bound $p_*(B) \le s_\tau(B)^2\|B\|$. For every admissible decomposition, subadditivity gives $p_*(H) \le \sum_a p_*(B_a) \le \sum_a s_\tau(B_a)^2\|B_a\|$, and taking the infimum yields $p_*(H) \le \mathcal{B}_\tau(H)$. Conversely the one-term decomposition shows $\mathcal{B}_\tau$ itself obeys the one-term bound. Hence $\mathcal{B}_\tau$ is the pointwise largest seminorm with that property, and the same argument applies to $\mathcal{B}$ and $\mathcal{T}_2$ with their own one-term costs.

Lower bounds by duality. Proving that a decomposition is optimal requires a certificate, and duality supplies one. Let $\mathcal{V}$ be the finite-dimensional real vector space of globally U(1)-invariant Hermitian operators, let $w(B,X)$ denote any of $s(B)^2\|B\|$, $s_\tau(B)^2\|B\|$ or $\tau(B)^2\|B\|$, and let p_w be the corresponding seminorm. Then

$$p_w(H) = \max_{\ell\in\mathcal{V}^*}\big\{\ell(H) : |\ell(B)| \le w(B,X) \text{ for every admissible labelled term } (B,X)\big\}. \tag{S33}$$

A feasible linear functional ℓ therefore certifies a lower bound on the cost, term by term, for every decomposition at once.

Proof. Any feasible ℓ bounds each decomposition termwise, so $\ell(H) \le p_w(H)$. Conversely, if $p_w(H) > 0$, the functional $f(aH) = a\,p_w(H)$ on the line $\mathbb{R}H$ is dominated by the sublinear functional p_w; the real Hahn–Banach theorem extends it to $\ell \in \mathcal{V}^*$ with $\ell(K) \le p_w(K)$ for all K, and applying this to $-K$ gives $|\ell(K)| \le p_w(K)$. The one-term decomposition gives $p_w(B) \le w(B,X)$, so the extension is feasible and attains $\ell(H) = p_w(H)$. If $p_w(H) = 0$ the zero functional suffices. □

Because Eq. (S33) never refers to a minimizing decomposition, a lower certificate does not require the infimum to be attained.

The XY ring, evaluated exactly. Let $L \ge 3$, let every site carry spin 1/2, and take the hopping Hamiltonian

$$H_{\text{hop}} = \sum_j \big(t_j S_j^+ S_{j+1}^- + \overline{t_j} S_j^- S_{j+1}^+\big). \tag{S34}$$

Let Z be Hermitian, globally U(1) invariant, of vanishing cost $\mathcal{T}_2(Z) = 0$, and with no nearest-neighbour off-diagonal matrix elements in the one-magnon sector. Then

$$\mathcal{T}_2(H_{\text{hop}} + Z) = \sum_j |t_j|. \tag{S35}$$

The cost is carried entirely by the hopping amplitudes; whatever else is present contributes nothing.

Proof of the lower bound. Write $|j\rangle$ for the state with a single up spin at site j. For an admissible B on X, build a graph G_B on X joining $u \ne v$ when $\langle u|B|v\rangle \ne 0$. Charge conservation prevents one-magnon edges from leaving X, and each edge contributes the transfer row $e_u - e_v$; hence any conserved weight is constant on each nontrivial component ω of G_B, and dropping the remaining constraints gives

$$\tau(B) \ge \sum_\omega \delta(\omega), \qquad \delta(\omega) = \min_c \sum_{j\in\omega} \operatorname{dist}_L(j,c), \tag{S36}$$

where δ is the circular dispersion about the best centre. Let $m(\omega)$ count nearest-neighbour ring edges with both ends in ω. If ω is a proper subset then $m(\omega) \le |\omega| - 1$, and packing $|\omega|$ distinct sites as close as possible around a centre gives distances $0, 1, 1, \ldots$ for odd $|\omega|$ and $\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \ldots$ for even $|\omega|$, both summing to

$$\delta(\omega) \ge \left\lfloor \frac{|\omega|^2}{4} \right\rfloor \ge |\omega| - 1. \tag{S37}$$

If ω is the whole ring then $m(\omega) = L$ and $\delta(\omega) = \lfloor L^2/4 \rfloor$, whose square is at least L. Either way

$$\sum_\omega m(\omega) \le \sum_\omega \delta(\omega)^2 \le \Big(\sum_\omega \delta(\omega)\Big)^2 \le \tau(B)^2, \tag{S38}$$

and since every matrix element is bounded by $\|B\|$,

$$\sum_j |\langle j|B|j{+}1\rangle| \le \tau(B)^2 \|B\|. \tag{S39}$$

Set $\phi_j = t_j/|t_j|$ when $t_j \ne 0$ and $\phi_j = 0$ otherwise, and define $\ell_t(K) = \sum_j \operatorname{Re}[\overline{\phi_j}\langle j|K|j{+}1\rangle]$. Equation (S39) makes ℓ_t feasible in Eq. (S33); the hypothesis on Z gives $\ell_t(Z) = 0$, while $\ell_t(H_{\text{hop}}) = \sum_j |t_j|$. The upper bound follows from subadditivity and the bond decomposition, each bond having transport distance one and norm $|t_j|$. □

The same evaluation at finite angle. For $L \geq 6$ the sharper quantity can also be computed. Let A be the one-magnon restriction of an admissible B and let F be the set of ring edges j with $\langle j|B|j{+}1\rangle \neq 0$. If F is nonempty, let R_B be the Hermitian signed edge matrix carrying the conjugate phase of $\langle j|B|j{+}1\rangle$ on each oriented edge. Compression gives $\|A\| \leq \|B\|$, the two orientations give $\mathrm{Tr}(R_B A) = 2\sum_j |\langle j|B|j{+}1\rangle|$, and Schatten 1–$\infty$ duality yields

$$\sum_j |\langle j|B|j{+}1\rangle| = \tfrac{1}{2}\,\mathrm{Tr}(R_B A) \leq g(F)\|B\|, \qquad g(F) = \tfrac{1}{2}\|R_B\|_*. \tag{S40}$$

If F is not the whole ring, its runs of consecutive edges have lengths $r_1, \ldots, r_c$; a run of r_i edges is unitarily equivalent, after gauging away the phases along it, to the adjacency matrix of a path on $r_i + 1$ vertices, whose eigenvalues are $2\cos[k\pi/(r_i+2)]$. Hence

$$g(F) = \sum_{i=1}^{c} \sum_{k=1}^{r_i+1} \Big|\cos\frac{k\pi}{r_i+2}\Big|, \qquad D(F) = \sum_{i=1}^{c} \Big\lfloor \frac{(r_i+1)^2}{4} \Big\rfloor, \tag{S41}$$

and the gauge freedom is why aligned edge phases do not enter. For the full ring set $D(F) = \lfloor L^2/4 \rfloor$. Distinct runs occupy disjoint vertex sets and merging them cannot lower the summed dispersions, so Eq. (S36) gives $\tau(B) \geq D(F)$, while subadditivity of the nuclear norm over single edges gives $g(F) \leq |F| \leq D(F)$.

Put $x = \pi/L$. If $D(F) < L/2$ then $D(F)\sin^2 x \leq \sin^2(D(F)x)$: the cases $D = 1, 2$ are direct, and for $D \geq 3$ the sine chord gives $\sin(Dx) \geq (2D/\pi)\sin x \geq \sqrt{D}\sin x$. If $D(F) \geq L/2$ the cost is clipped at one, and it remains to check that the left side cannot exceed it. For $L \geq 7$, with $m = |F|$ edges and at most $n \leq L$ incident vertices, the Frobenius bound gives $g(F) \leq \sqrt{mn/2} \leq L/\sqrt{2}$ and hence $g(F)\sin^2(\pi/L) \leq \pi^2/(\sqrt{2}L) < 1$. For $L = 6$: if $m \leq 4$ then $g(F) \leq m \leq 4$; if $m = 5$ the six-vertex path has $g(F) = \csc(\pi/14) - 1 < 4$; and if $m = 6$, pairing antipodal ring eigenvalues gives $g(F) = 2\sum_{k=0}^{2}|\cos(\theta + k\pi/3)| \leq 4$. Since $\sin^2(\pi/6) = 1/4$, the endpoint holds too. Therefore

$$\sin^2\frac{\pi}{L} \sum_j |\langle j|B|j{+}1\rangle| \leq s_\tau(B)^2\|B\|, \tag{S42}$$

so $\sin^2(\pi/L)\,\ell_t$ is feasible in the finite-angle version of Eq. (S33). Its value on $H_{\mathrm{hop}} + Z$ is $\sin^2(\pi/L)\sum_j |t_j|$, and since $\mathcal{B}_\tau(Z) \leq (2\pi S/L)^2 \mathcal{T}_2(Z) = 0$ the bond decomposition matches it. Hence

$$\mathcal{B}_\tau(H_{\mathrm{hop}} + Z) = \sin^2\frac{\pi}{L}\sum_j |t_j|, \qquad L \geq 6. \tag{S43}$$

For the uniform XY ring, $t_j = J/2$ gives $\mathcal{T}_2 = L|J|/2$. Presented instead as a single ring-wide term the same Hamiltonian has $\tau = \lfloor L^2/4 \rfloor$ and costs at least $|J|\lfloor L^2/4\rfloor^2$. The bond decomposition therefore attains the infimum and removes a genuine inflation that has nothing to do with the physics. More generally, if p divides L, $t_{j+p} = t_j$, and Z also commutes with T^p, then for pN/L not an integer

$$\Delta_N(H_{\mathrm{hop}} + Z) \leq 2\sin^2\frac{\pi}{L}\sum_j |t_j| = \frac{2L}{p}\sin^2\frac{\pi}{L}\sum_{a=0}^{p-1}|t_a|. \tag{S44}$$

For real p-periodic exchange and Dzyaloshinskii–Moriya couplings, $J_j(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y) + D_j(S_j^x S_{j+1}^y - S_j^y S_{j+1}^x)$ corresponds to $t_j = (J_j + iD_j)/2$, which proves the unit-cell bound of the main text for dimerized chains with Dzyaloshinskii–Moriya coupling. At $p = 2$ and half filling $pN/L = 1$, so the hypothesis fails and the corollary correctly makes no claim about the dimerization gap.

VIII. TRANSFER MATRICES AND THE TRANSPORT DISTANCE

This section turns $\tau(B)$ from a definition into something one evaluates, gives its closed form for the processes that occur in practice, and shows why the infimum over decompositions in Eq. (S2) cannot be dispensed with.

Let $|a\rangle$ be the product S^z basis and $M(c) = \sum_j c_j S_j^z$. Its commutator has matrix elements

$$\langle a|[B, M(c)]|b\rangle = \langle a|B|b\rangle \sum_j c_j[m_j(b) - m_j(a)], \tag{S45}$$

so $[B, M(c)] = 0$ exactly when c is orthogonal to every charge transfer the term actually performs. Collecting those transfers as the rows of a matrix D_B, and writing $x = (j)_{j\in X}$ for the site positions,

$$\mathcal{K}_X(B) = \ker_{\mathbb{R}} \mathsf{D}_B, \qquad \tau(B) = \inf_{n\in\mathbb{Z}^X} \operatorname{dist}_1\big(x + Ln, \ker \mathsf{D}_B\big). \tag{S46}$$

The transport distance is thus the ℓ^1 distance from the site positions to the space of conserved weights, minimized over ways of counting sites around the seam. Since exact zeros of D_B can be read off a term by inspection, this is a certificate rather than a numerical search.

Lower bounds come from duality. For a subspace K of a finite-dimensional real space with the ℓ^1 norm, the dual of the quotient is the annihilator $K^\perp$ with the restricted ℓ^∞ norm, so

$$\operatorname{dist}_1(v, K) = \max_{y\in K^\perp,\ \|y\|_\infty\le 1} y\cdot v, \tag{S47}$$

and $(\ker \mathsf{D}_B)^\perp = \operatorname{ran} \mathsf{D}_B^T$ identifies the witnesses as combinations of the transfer rows themselves.

Closed form for a fixed transfer pattern. Suppose the nonzero integer rows of D_B span a one-dimensional real space with primitive generator $d \in \mathbb{Z}^X$, unique up to sign with $\gcd_j |d_j| = 1$. Then

$$\tau(B) = \frac{\operatorname{dist}(d\cdot x,\ L\mathbb{Z})}{\|d\|_\infty}. \tag{S48}$$

The term costs what its charge displacement $d \cdot x$ falls short of a whole number of turns around the ring.

Proof. Here $\ker \mathsf{D}_B = \ker d$, and Eq. (S47) gives $\operatorname{dist}_1(v, \ker d) = |d\cdot v|/\|d\|_\infty$ for each fixed lift $v = x + Ln$. Primitivity and Bézout's identity make $\{d\cdot n : n \in \mathbb{Z}^X\} = \mathbb{Z}$, so minimizing over n moves $d\cdot x$ to the nearest multiple of L. Shifting a representative by L changes $d\cdot x$ by a multiple of L, and flipping the sign of d changes nothing. □

For a spin-1/2 term with fixed transfer, the transfer vector $d \in \{-1, 0, 1\}^X$ is automatically primitive and the Hermitian conjugate supplies $-d$, so

$$\tau(B) = \operatorname{dist}\Big(\sum_{j\in X} d_j j,\ L\mathbb{Z}\Big), \tag{S49}$$

which vanishes exactly when the wrapped polarization displacement is a multiple of the circumference. This covers ordinary dipole-neutral processes and processes that wrap the seam alike. At higher spin an actual transfer may be a non-primitive multiple of d, and its finite twist phase can then cancel even when Eq. (S48) is positive — Eq. (S22) is such a case. The formula concerns the real span of the transfer rows, which is what defines τ, not the integer lattice they generate.

Non-additivity, and the need for the decomposition infimum. The cost of a single term is not additive, so a Hamiltonian must be allowed to choose how it is written. Take $L = 6$, sites $x = (0, 1, 3, 4)$, and two Hermitian spin-1/2 fixed-transfer terms with

$$d_1 = (-1, -1, 1, 1), \qquad d_2 = (-1, 1, 1, -1). \tag{S50}$$

Their displacements are 6 and 0, so by Eq. (S48) each term is free. The transfer kernel of the *sum*, however, is $\{(a, b, a, b)\}$, and for every integer lift $v = x + 6n$,

$$\operatorname{dist}_1(v, \ker \mathsf{D}) = |v_0 - v_2| + |v_1 - v_3| \ge 3 + 3 = 6, \tag{S51}$$

with equality attained — the sum, taken as one term, costs 6. Keeping the two summands as separate admissible terms instead proves that all three Hamiltonian costs of the sum vanish. A specified term with a specified lift is certified by Eq. (S46); optimizing over all decompositions is a separate algorithmic question, and any explicit decomposition already gives a rigorous upper bound on the seminorm and hence on the gap.

Finally, if the transfer rows are the edge vectors $e_i - e_j$ of a graph, a weight lies in the kernel exactly when it is constant on each component. Integer lifts and constants then separate componentwise, and minimizing their ℓ^1 costs gives the sum of circular dispersions used in Eq. (S36).

IX. REDUCTION TO FINITE-RANGE GEOMETRY

Replacing the conserved weights in the definition (5) of the transport distance by constant vectors — the crudest admissible choice — turns transport back into geometry and reproduces the standard finite-range theorem with an explicit constant. This section carries out that reduction, and so shows that Result B is never weaker than the familiar estimate.

For a set of sites X define its circular dispersion

$$\delta(X) = \min_c \sum_{j \in X} \mathrm{dist}_L(j, c), \tag{S52}$$

the smallest total distance from the sites of X to a common centre on the ring; taking c constant in the definition of τ gives $\tau(B) \le \delta(X)$ whenever B is supported on X.

Suppose every support has cycle diameter at most R and the local strength is bounded,

$$\sup_x \sum_{X \ni x} \|\Phi_X\| \le J. \tag{S53}$$

If $L > 2R$, anchor X at one of its sites and give every other site its unique principal displacement in $[-R, R]$; distinct sites take distinct values, so the worst case uses every nonzero integer there and $\delta(X) \le R(R+1)$. If $L > 3R$ the support lifts to an integer interval of diameter at most R, and among at most one site per position the sum of absolute deviations about a median is largest when every position is occupied,

$$\max_{Y \subset \{0,\dots,R\}} \min_c \sum_{j \in Y} |j - c| = \left\lfloor \frac{(R+1)^2}{4} \right\rfloor = \kappa_R, \tag{S54}$$

with equality for the full consecutive set. Summing Eq. (S53) over sites gives $\sum_X |X| \|\Phi_X\| \le LJ$, and since every term has $|X| \ge 1$ and the charge average contracts norms, $\sum_X \|B_X\| \le LJ$. Combining with Result A, for every N with pN/L not an integer,

$$\Delta_N(H_L) \le \frac{8\pi^2 S^2 R^2 (R+1)^2 J}{L} \quad (L > 2R), \qquad \Delta_N(H_L) \le \frac{8\pi^2 S^2 \kappa_R^2 J}{L} \quad (L > 3R). \tag{S55}$$

Both are the familiar $O(1/L)$, now with the geometric constant displayed. Result B is sharper whenever the interaction conserves weighted charges beyond the total, since then the constant-vector choice is not optimal.

X. LONG-RANGE DECAY CLASSES

Specialize to one site per cell, $p = 1$; all bounds hold for every N with N/L not an integer. The point of this section is that the two thresholds known in the literature are the two ways transport can behave as the range grows, and both follow from the same functional.

For one translation orbit of two-site exchange terms at each cycle distance r, with $\|B_{j,r}\| \le C(1+r)^{-\alpha}$, the exchange moves charge the full distance, $\tau(B_{j,r}) \le r$, and Result B gives

$$\Delta_N(H_L) \le \frac{8\pi^2 S^2 C}{L} \sum_{r=1}^{\lfloor L/2 \rfloor} \frac{r^2}{(1+r)^\alpha}. \tag{S56}$$

This is $O(L^{-1})$ for $\alpha > 3$, $O(L^{-1}\log L)$ at $\alpha = 3$, and $O(L^{2-\alpha})$ for $2 < \alpha < 3$; in every case it vanishes for $\alpha > 2$. The two powers of r are the cost of transporting charge across the term.

If instead the transport distance stays bounded, $\tau(B_{j,r}) \le K$ uniformly in j, r and L — as for a product of two separately neutral clusters of bounded internal spread — those two powers disappear and

$$\Delta_N(H_L) \le \frac{8\pi^2 S^2 C K^2}{L} \sum_{r=1}^{\lfloor L/2 \rfloor} \frac{1}{(1+r)^\alpha}, \tag{S57}$$

which is $O(L^{-1})$ for $\alpha > 1$, $O(L^{-1}\log L)$ at $\alpha = 1$, and $O(L^{-\alpha})$ for $0 < \alpha < 1$, so it vanishes for every $\alpha > 0$. Equations (S56) and (S57) are the one-dimensional Type-I and Type-II thresholds of Ref. [S1], recovered here as the two regimes of a single quantity.

More generally, for any sequence of rings and any chosen sequence of fractional sectors,

$$\mathcal{T}_2(H_L) = o(L^2) \implies \Delta_{N_L}(H_L) \to 0. \tag{S58}$$

XI. DIRECTION-RESOLVED BOUNDS ON CYLINDERS AND LADDERS

Twisting one direction of a cylinder charges an interaction only for the displacement it produces along that direction. This section proves the directional bound, shows that intra-slice dynamics is exactly free, and works out the width–decay crossover it implies.

Take L slices around a periodic direction, each slice a copy of a finite graph with W sites, and let each site carry a finite-dimensional charge with integer spectrum. Twist only the periodic coordinate,

$$U^{\parallel} = \exp\Big(\frac{2\pi i}{L}\sum_x \sum_a x\, q_{x,a}\Big), \tag{S59}$$

where x indexes slices and a indexes sites within a slice. With T advancing x and preserving a, integer charge gives exactly as before

$$T^p U^{\parallel} T^{-p} = U^{\parallel} e^{-2\pi i p Q/L}. \tag{S60}$$

Assume p divides L, $[H_L, Q] = 0$ and $[H_L, T^p] = 0$. Now repeat Sec. IV verbatim, with each charge assignment carrying the longitudinal phase

$$\theta^{\parallel}_{\lambda} = \frac{2\pi}{L}\sum_{(x,a)\in X} x\,\lambda_{x,a} \pmod{2\pi}, \tag{S61}$$

which counts only how far along the ring the charge sits, ignoring where it lies within a slice. This defines a directional spread $\eta_{\parallel}(B)$, a directional term cost $s_{\parallel}(B) = \sin(\min\{\eta_{\parallel}(B), \pi/2\})$, and a directional Hamiltonian cost $\mathcal{B}_{\parallel}$ by the same infimum. Then for every N with pN/L not an integer,

$$\Delta_N(H_L) \le 2\mathcal{B}_{\parallel}(H_L). \tag{S62}$$

Proof. Apply the charge average of Sec. II to any supported Hermitian decomposition; it contracts norms and produces $[B_X, Q_X] = 0$, and it commutes with conjugation by the local longitudinal twist since both are generated by the same commuting charges, so a twist-invariant term stays invariant. The exact-logarithm theorem and Eq. (S23) then give

$$\Big\|\tfrac{1}{2}\big((U^{\parallel})^{-1}B_X U^{\parallel} + U^{\parallel}B_X(U^{\parallel})^{-1}\big) - B_X\Big\| \le 2s_{\parallel}(B_X)^2\|B_X\|. \tag{S63}$$

By Eq. (S60) each of $(U^{\parallel})^{\pm 1}\Psi$ is orthogonal to a unique sector ground state Ψ, though the two twisted states need not be orthogonal to each other. Averaging their excitation energies, applying the sector variational principle, and taking the decomposition infimum gives Eq. (S62); a degenerate sector ground energy makes it automatic. The same proof works for any chosen periodic direction of a finite product torus, one direction at a time; it does not combine several twists into a higher-dimensional flux-insertion theorem. $\square$

Exact vanishing of the intra-slice cost. If B is supported inside the single slice $x = x_0$, its local longitudinal twist is $e^{2\pi i x_0 Q_X/L}$, a function of the charge on X alone, and so commutes with B. Hence

$$\eta_{\parallel}(B) = 0. \tag{S64}$$

Arbitrary charge-conserving transverse hopping, intra-slice many-body interaction, and intra-slice disorder therefore cost nothing along the ring, whatever their strength, order, or transverse range; so do product-charge diagonal interactions of arbitrary longitudinal range.

Selective hopping. Let every site carry integer levels $0,\ldots,M$ with $M \ge 1$. Choose one oriented representative e of each Hermitian hopping bond, running from a site in one slice to a site r_e slices along, and write

$$H_L = \sum_e \big(t_e|1,0\rangle\langle 0,1|_e + \overline{t_e}|0,1\rangle\langle 1,0|_e\big) + V, \tag{S65}$$

where the bonds and amplitudes are p-periodic, $[V,Q] = [V,T^p] = 0$, and V decomposes into charge-conserving terms commuting with their own local longitudinal twists. Writing ℓ_e for the distance from r_e to the nearest multiple of L, the edge term has norm $|t_e|$ and directional spread $\pi\ell_e/L$, so Eq. (S62) gives

$$\Delta_N(H_L) \le 2\sum_e |t_e|\sin^2\frac{\pi\ell_e}{L}, \qquad pN/L \notin \mathbb{Z}. \tag{S66}$$

Nothing here depends on transverse separation, on the unused charge levels, or on any admitted term of V, and edges with $\ell_e = 0$ contribute exactly zero. For the fiber-preserving nearest-neighbour subclass with $t_{x,a} = t_{c,a}$, $c = x \bmod p$,

$$\Delta_N(H_L) \le \frac{2L}{p}\sin^2\frac{\pi}{L}\sum_a\sum_{c=0}^{p-1}|t_{c,a}|, \qquad \text{and for uniform } t, \qquad \Delta_N(H_L) \le 2LW|t|\sin^2\frac{\pi}{L} \le \frac{2\pi^2 W|t|}{L}. \tag{S67}$$

The bound vanishes for bounded hopping whenever $W = o(L)$. At isotropic aspect ratio $W \asymp L$ this uniform-twist estimate is only $O(1)$ and makes no two-dimensional claim. At $W = 1$ it returns the multilevel chain, and its independence of M removes the spurious factor M^2 that the charge-width certificate of Eq. (S19) produces in the quadratic relaxation.

Power-law channels. Write $n = \lfloor L/2 \rfloor$ and let

$$A_L(r) = \sum_{e:\,\ell_e = r}|t_e| \tag{S68}$$

be the total amplitude moving charge exactly r slices. Grouping Eq. (S66) by r gives the exact profile estimate $\Delta_N \le 2\sum_r A_L(r)\sin^2(\pi r/L)$, and assuming $A_L(r) \le c_0 L K_L r^{-\alpha}$ with $\alpha > 2$ and a channel mass K_L, the inequality $\sin x \le x$ yields

$$\Delta_N(H_L) \le \frac{2\pi^2 c_0 K_L}{L}\sum_{r=1}^{n} r^{2-\alpha}, \qquad \sum_{r=1}^{n} r^{2-\alpha} \le \begin{cases} \zeta(\alpha-2), & \alpha > 3, \\ 1 + \log n, & \alpha = 3, \\ 1 + \dfrac{n^{3-\alpha}-1}{3-\alpha}, & 2 < \alpha < 3. \end{cases} \tag{S69}$$

For $K_L = O(L^\beta)$ the three branches scale as $O(L^{\beta-1})$, $O(L^{\beta-1}\log L)$ and $O(L^{\beta+2-\alpha})$, so the bound vanishes throughout

$$\beta < \min\{1, \alpha - 2\}, \tag{S70}$$

with $K_L = o(L/\log L)$ at the dipolar boundary $\alpha = 3$. The hypothesis $K_L = O(W)$ is a statement about the amplitude that actually moves charge between slices; it holds, for instance, when each distance and longitudinal cell carries $O(W)$ moving bonds with bounded prefactors. Bounded degree of the slice graph is not sufficient, since a dense inter-slice graph can carry a larger channel mass — while purely transverse charge-conserving dynamics remains exactly free.

Dense Euclidean strip. Take $p = 1$, let the slice be a ring of $W \le n$ sites, and allow every pair of transverse channels to couple. For one representative of each Hermitian longitudinal bond let $r_e \ge 1$ be the longitudinal and $s_e \ge 0$ the transverse cycle distance, and assume the Euclidean tail

$$|t_e| \le J(r_e^2 + s_e^2)^{-\alpha/2}, \qquad \alpha > 2. \tag{S71}$$

Let $\nu_W(s) = 1$ if W is even and $s = W/2$, and $\nu_W(s) = 2$ otherwise, so that $1 + \sum_s \nu_W(s) = W$ counts the transverse antipode once. When L is even, $r = L/2$ is represented by the half longitudinal orbit and translation covariance forces the corresponding channel matrix to be Hermitian, with self-reversing simultaneous-antipode entries real, so each physical bond is counted exactly once. Counting longitudinal translates and transverse sources,

$$A_{L,W}(r) \le JLW\Big[r^{-\alpha} + \sum_{s=1}^{\lfloor W/2 \rfloor}\nu_W(s)(r^2 + s^2)^{-\alpha/2}\Big]. \tag{S72}$$

With

$$c_\alpha = 1 + \frac{\sqrt{\pi}\,\Gamma((\alpha-1)/2)}{\Gamma(\alpha/2)}, \tag{S73}$$

the transverse summand is decreasing and comparison with its integral over the half-line gives two complementary estimates for the bracket, namely $\le c_\alpha r^{1-\alpha}$ and $\le W r^{-\alpha}$. Using the first for $r \le W$ and the second for $r > W$, then $\sin x \le x$,

$$\Delta_N(H_L) \le \frac{2\pi^2 JW}{L}\Big[c_\alpha\sum_{r=1}^{W} r^{3-\alpha} + W\sum_{r=W+1}^{n} r^{2-\alpha}\Big]. \tag{S74}$$

Integral comparison gives the explicit branches

$$\Delta_N(H_L) \leq \begin{cases} \dfrac{2\pi^2 JW}{L}\Big[c_\alpha \zeta(\alpha-3) + \dfrac{1}{\alpha-3}\Big], & \alpha > 4, \\ \dfrac{2\pi^2 JW}{L}\big[(1+\pi/2)(1+\log W) + 1\big], & \alpha = 4, \\ \dfrac{2\pi^2 JW^{5-\alpha}}{L}\Big[\dfrac{c_\alpha}{4-\alpha} + \dfrac{1}{\alpha-3}\Big], & 3 < \alpha < 4, \\ \dfrac{2\pi^2 JW^2}{L}\Big[3 + \log\dfrac{n}{W}\Big], & \alpha = 3, \\ \dfrac{2\pi^2 JW^2}{L} n^{3-\alpha}\Big[c_\alpha + \dfrac{1}{3-\alpha}\Big], & 2 < \alpha < 3. \end{cases} \tag{S75}$$

At the dipolar endpoint the constant is not hidden inside an asymptotic statement: $c_3 = 1 + \sqrt{\pi}\,\Gamma(1)/\Gamma(3/2) = 3$, and $\sum_{r=W+1}^{n} r^{-1} \leq \log(n/W)$, so the near part contributes $3W$ and the far part at most $W\log(n/W)$ inside the bracket. This is the dipolar strip bound quoted in the main text, and it vanishes whenever $W^2[1+\log(L/W)] = o(L)$; a charge-moving van der Waals tail gives $O(W/L)$. For $W = O(L^\beta)$ the strict width thresholds are

$$\beta < \begin{cases} (\alpha-2)/2, & 2 < \alpha \leq 3, \\ 1/(5-\alpha), & 3 \leq \alpha < 4, \\ 1, & \alpha \geq 4, \end{cases} \tag{S76}$$

with logarithmic refinements at $\alpha = 3, 4$. This is a sufficient anisotropic region, not a converse and not an isotropic two-dimensional theorem. Dipolar spin exchange in polar-molecule arrays supplies the relevant charge-conserving mechanism [S2, S3, S4], though we do not assert that those experiments realize the periodic strip geometry, filling, or spectral hypotheses assumed here. A coherently driven Rydberg Ising simulator [S5], by contrast, does not conserve Rydberg occupation and is not covered merely because its density interaction is van der Waals.

Odd-leg spin-1/2 ladder. Let L be even and W odd, and take

$$H_L = \sum_x \sum_{a=1}^{W} J_a\, \boldsymbol{S}_{x,a}\cdot\boldsymbol{S}_{x+1,a} + \sum_x h_x^{\mathrm{rung}} + Z, \tag{S77}$$

with real J_a, the h_x^{rung} translates of an arbitrary SU(2)-invariant Hermitian operator supported in one rung, and Z translation and SU(2) invariant with a decomposition into terms commuting with their local longitudinal twists. No norm or interaction-order bound is placed on h_x^{rung} or on the admitted terms of Z. Using $q_{x,a} = S_{x,a}^z + 1/2$, every rung term is longitudinally free by Eq. (S64), and the charge-moving part of a longitudinal Heisenberg bond has norm $|J_a|/2$ and spread π/L. If the ground state is degenerate the full gap is zero; otherwise SU(2) invariance makes it a singlet with $N = LW/2$, and W odd with L even makes N/L non-integer. Hence

$$\Delta(H_L) \leq L\sin^2\frac{\pi}{L}\sum_{a=1}^{W}|J_a| \leq \frac{\pi^2}{L}\sum_{a=1}^{W}|J_a|, \tag{S78}$$

which for uniform longitudinal exchange is $\pi^2 W|J_\parallel|/L$ and vanishes for bounded $J_\parallel$ whenever $W = o(L)$. If instead the longitudinal exchanges are long ranged with $\sum_a |J_{a,r}| \leq CWr^{-\alpha}$ at distance r, the same counting gives

$$\Delta(H_L) \leq L\sum_{r,a}|J_{a,r}|\sin^2\frac{\pi r}{L} \leq \frac{\pi^2 CW}{L}\sum_{r=1}^{n} r^{2-\alpha}. \tag{S79}$$

Fixed-width ladder statements are known [S6, S7], as are higher-dimensional bounds on tori [S8]; what is recorded here is the direction-resolved finite-volume cost, the crossover as the width grows, and exact independence from arbitrary admitted rung dynamics.

XII. THE SPIN-1/2 MODEL WITH ZERO-COST TAILS

Here we derive the bound quoted in the main text for that model, with every finite-size factor explicit, and verify that both advertised classes of free interaction really are free.

Write the transverse bond of the model as

$$B_j = J_\perp\big(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y\big) = \frac{J_\perp}{2}\big(S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+\big). \tag{S80}$$

On the span of $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$ it is the off-diagonal matrix with entries $J_\perp/2$ and it annihilates the other two product states, so $\|B_j\| = |J_\perp|/2$. For $J_\perp \neq 0$ its transfer graph is the single edge $(j, j{+}1)$ and $\tau(B_j) = 1$; otherwise $\tau(B_j) = 0$.

Diagonal tails. Any operator diagonal in the product S^z basis commutes with $\sum_j c_j S_j^z$ for *every* real c, so its conserved-weight space is all of $\mathbb{R}^X$ and the infimum in τ is zero. Hence $\tau(V) = 0$ even when the whole diagonal part V is retained as a single ring-wide term, and no decay condition on its couplings is needed anywhere.

Polarization-neutral hoppings. Write $Z = \sum_\gamma B_\gamma$, keeping each Hermitian fixed-transfer spin-1/2 monomial as a separate admissible term. If its transfer vector satisfies $\sum_j d_j^{(\gamma)} j \in L\mathbb{Z}$ then Eq. (S49) gives $\tau(B_\gamma) = 0$. A nontrivial zero-cost spin-1/2 transfer cannot consist of one raising and one lowering operator, since that is a transfer between two distinct ring sites and has nonzero cycle distance; every such B_γ therefore carries at least two raising and two lowering operators and annihilates the one-magnon sector. The displayed decomposition thus proves

$$\mathcal{T}_2(Z) = \mathcal{B}_\tau(Z) = \mathcal{B}(Z) = 0, \tag{S81}$$

even though $\tau(Z)$ computed for the sum as one term need not vanish — the non-additivity of Eq. (S50) made explicit.

For the counterflow term of the main text, use the integer lifts $(j,\, j{+}1,\, j{+}r,\, j{+}r{+}1)$, which are valid even when either bond crosses the seam. The first monomial has transfer vector $d = (1, -1, -1, 1)$ and its conjugate has $-d$, and

$$d \cdot (j,\, j{+}1,\, j{+}r,\, j{+}r{+}1) = 0. \tag{S82}$$

The two hops move charge in opposite senses by equal amounts, so the term commutes with the weighted charge whose coefficients are those lifts, and choosing that same pair in the definition of τ gives $\tau(C_{j,r}) = 0$ exactly. For $g_r \neq 0$ the term has nonzero off-diagonal matrix elements in the product basis; when L is even and $r = L/2$, translation by r exchanges the two monomials, so the orbit sum depends only on $\operatorname{Re} g_r$, and at least one non-cancelling orbit produces a genuine off-diagonal interaction.

The exact cost and the resulting bound. Neither V nor Z has one-magnon nearest-neighbour matrix elements, so with $H_L = H_{\mathrm{hop}} + V + Z$ the hypotheses of Eq. (S35) hold and

$$\mathcal{T}_2(H_L) = \frac{L|J_\perp|}{2}. \tag{S83}$$

Result B evaluated on the bond decomposition already gives the bound of the main text. It can also be read directly off the exact twist response, which is instructive because no inequality is used until the last line. Put $\theta = 2\pi/L$. Diagonality of V and the displacement condition on Z make the uniform-twist phase of every displayed constituent cancel exactly, while the two hopping directions acquire opposite phases, so

$$\frac{U^{-1} H_{\mathrm{hop}} U + U H_{\mathrm{hop}} U^{-1}}{2} = \cos\theta\, H_{\mathrm{hop}}, \qquad [V + Z, U] = 0. \tag{S84}$$

If the fractional-sector ground state Ψ is unique, the excitation energies $e_\pm$ of $U^{\pm 1}\Psi$ are nonnegative and their trial states are orthogonal to Ψ by Sec. VI, so

$$0 \le \frac{e_+ + e_-}{2} = (\cos\theta - 1)\langle \Psi, H_{\mathrm{hop}}\Psi\rangle \le (1 - \cos\theta)\|H_{\mathrm{hop}}\| \le \frac{L|J_\perp|}{2}(1 - \cos\theta). \tag{S85}$$

One twist sign lies below the average, and the sector variational principle, together with the automatic zero gap in the degenerate case, gives

$$\Delta_N(H_L) \le L|J_\perp| \sin^2\frac{\pi}{L} \le \frac{\pi^2 |J_\perp|}{L}, \qquad N/L \notin \mathbb{Z}. \tag{S86}$$

The first form is strictly sharper than the quadratic one whenever $J_\perp \neq 0$, and Result C converts it into a full-gap bound when its spectral condition holds.

Comparison with a support-based estimate. For comparison, pricing the diagonal tail by its diameter assigns $r^2\|V_r S^z_j S^z_{j+r}\| = r^2|V_r|/4$ to each of the L terms at distance r; multiplying by the spin-1/2 prefactor $2\pi^2/L^2$ gives an extra contribution

$$\frac{\pi^2}{2L}\sum_{r=1}^{\lfloor (L-1)/2\rfloor} r^2|V_r|, \tag{S87}$$

which for $V_r = V/r^\alpha$ is the expression quoted in the main text. The restriction $r \le \lfloor (L-1)/2\rfloor$ makes r the unambiguous shortest cycle distance; a general diagonal V may also contain the even-ring antipodal term. The true contribution of all of it is zero.

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